

## Analysis First-Year Exam, January 2023, Part 1

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Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let  $(a_n)$  and  $(b_n)$  be bounded real sequences. Prove that

$$\limsup(a_n + b_n) \leq \limsup a_n + \limsup b_n$$

with equality when  $(a_n)$  converges.

2. Let  $A$  be a subset of the metric space  $(M, d)$  and define  $d_A : M \rightarrow \mathbb{R}$  by  $d_A(x) = \inf\{d(a, x) : a \in A\}$ . Prove that:

- (i)  $d_A$  is uniformly continuous;
- (ii)  $\{x \in M : d_A(x) = 0\} = \overline{A}$ .

3. Let  $X$  and  $Y$  be metric spaces; let  $f : X \rightarrow Y$  be a continuous bijection, with inverse  $g : Y \rightarrow X$ . For each of the following statements, provide a proof if true or a counterexample (with justification) if false:

- (i) if  $X$  is compact then  $g$  is continuous;
- (ii) if  $Y$  is compact then  $g$  is continuous.

4. Let  $X$  and  $Y$  be metric spaces; let  $f : X \rightarrow Y$  be a continuous bijection, with uniformly continuous inverse  $g : Y \rightarrow X$ . For each of the following statements, provide a proof if true or a counterexample if false:

[Be sure to demonstrate uniform continuity of  $g$  in any counterexample.]

- (i) if  $X$  is complete then  $Y$  is complete;
- (ii) if  $Y$  is complete then  $X$  is complete.

5. The function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is differentiable, with  $|f'| \leq 1$  everywhere. Prove that if  $\varepsilon > 0$  is sufficiently small (how small?) then the function  $F : \mathbb{R} \rightarrow \mathbb{R}$  defined by  $F(x) = x + \varepsilon f(x)$  is injective.